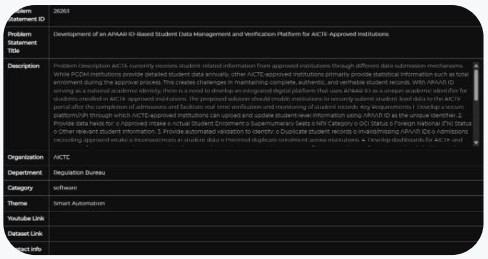
2026-10-26 13:04:22 **You**



Problem Statement ID	26263
Problem Statement Title	Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
Description	Problem Description AICTE currently receives student-related information from approved institutions through different data submission mechanisms. While PCON institutions provide detailed student data annually, other AICTE-approved institutions primarily provide statistical information such as total enrolment during the approval process. This creates challenges in maintaining complete, authentic, and verifiable student records. With APAAR ID serving as a national academic identity, there is a need to develop an integrated digital platform that uses APAAR ID as a unique academic identifier for students enrolled in AICTE-approved institutions. The proposed solution should enable institutions to securely submit student-level data to the AICTE portal after the completion of admissions and facilitate real-time verification and monitoring of student records. Key Requirements 1. Develop a secure platform/API through which AICTE-approved institutions can upload and update student-level information using APAAR ID as the unique identifier. 2. Provide data fields for: o Approved intake o Actual Student Enrolment o Supernumerary Seats o NRI Category o OCI Status o Foreign National (FN) Status o Other relevant student information. 3. Provide automated validation to identify: o Duplicate student records o Invalid/missing APAAR IDs o Admissions exceeding approved intake o Inconsistencies in student data o Potential duplicate enrolment across institutions 4. Develop dashboards for AICTE and
Organization	AICTE
Department	Regulation Bureau
Category	software
Theme	Smart Automation
Youtube Link	
Dataset Link	
Contact Info	

okay so here i have given you my research of the project to you that i have done till now and giving it to you so that u can make it much better refine it more suggest some more changes and betterments more practically possible professional and more applicable and problem solving solution you have to think like a lead architect and make this u know more resume efficient and a standard levelled project u have to analyze it and go through it deeply and research more about it generate every correction changes suggestions in a report simultaneously how to think this can be transformed into a one of the great projects industry relevant and efficient this should actually solve some problem which is listed i am giving u the actual problem statement along with the research done so work accordingly

Claude

I'll start by reading your two research files and the skill for building the report.

I'm realigning your v1/v2 research to the actual problem statement in your screenshot, so first I'll look for the rest of its text and the current rules it depends on.

The report is ready: https://claude.ai/artifact/5Hd3tienC9ipwJWXPgfNUu. Read it over, edit inline, or comment.

The main finding is that your v1 and v2 documents solve a different problem from the one in your screenshot.

- **What the problem statement asks for:** AICTE wants student-level data submitted with APAAR ID as the key. It names fields for approved intake, actual enrolment, supernumerary seats, NRI, OCI and Foreign National status. It also wants validation for duplicates, invalid or missing IDs, admissions above approved intake, inconsistencies, and duplicate enrolment across institutions, plus dashboards.

- **What v1 and v2 built:** deep work on marksheet forgery detection, OCR, credit transfer and verifiable credentials. None of the named data fields appears in either file, and the over-admission check is missing.
- **What v3 does:** it makes the problem statement's own checklist the core of the product. It keeps your best ideas (tenant isolation, no Aadhaar storage, signed and tamper-evident submissions, the live tamper demo) as a smaller trust layer. Forensics, OCR and credentials move to a roadmap slide.

The report also covers:

- a worked over-admission example, including a case where a naive intake check would raise a false alarm
- the validation engine and cross-institution duplicate detection
- architecture, API and dashboards
- a plan from now to the December finale, a demo script, and resume and interview material

What I couldn't confirm:

- **Full problem statement:** your screenshot cuts off after requirement 4 and the start of the dashboard line, so please paste the rest.
- **Seat caps:** the supernumerary percentages come from a news report on the 2026-27 handbook, and the NRI cap comes from a secondary source. Please check both against the handbook text.
- **SIH finale rules:** the finale dates and team rules come from secondary sources. Please check the official SIH site on whether you may bring pre-built code.
- **APAAR verification:** I found that ABC offers APIs for verifying APAAR IDs. Prototype access is unconfirmed, so the plan uses an adapter with a mock.

If you send the full statement and the fields AICTE currently collects from PGDM institutions, I can turn the rules, schema and API sections into a full specification.

2026-10-26 13:58:50 **You**

Problem Statement Title
Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
Description
Problem Description AICTE currently receives student-related information from approved institutions through different data submission mechanisms. While PGDM institutions provide detailed

student data annually, other AICTE-approved institutions primarily provide statistical information such as total enrolment during the approval process. This creates challenges in maintaining complete, authentic, and verifiable student records. With APAAR ID serving as a national academic identity, there is a need to develop an integrated digital platform that uses APAAR ID as a unique academic identifier for students enrolled in AICTE-approved institutions. The proposed solution should enable institutions to securely submit student-level data to the AICTE portal after the completion of admissions and facilitate real-time verification and monitoring of student records. Key Requirements 1. Develop a secure platform/API through which AICTE-approved institutions can upload and update student-level information using APAAR ID as the unique identifier. 2. Provide data fields for: o Approved Intake o Actual Student Enrolment o Supernumerary Seats o NRI Category o OCI Status o Foreign National (FN) Status o Other relevant student information. 3. Provide automated validation to identify: o Duplicate student records o Invalid/missing APAAR IDs o Admissions exceeding approved intake o Inconsistencies in student data o Potential duplicate enrolment across institutions. 4. Develop dashboards for AICTE and institutions for monitoring admissions and student records in real time. 5. Provide a secure verification mechanism for authorized stakeholders such as AICTE, institutions, employers, and other regulatory bodies. 6. Enable generation of verifiable student records linked to the student's academic identity. 7. Provide integration/API capability with existing institutional and AICTE systems. 8. Ensure appropriate data security, privacy, authentication, role-based access control, audit trails, and consent mechanisms. 9. Enable generation of reports for policy-making, accreditation, monitoring, and regulatory purposes. 10. Support printing/display of APAAR ID along with the student ID on certificates issued by degree-awarding bodies. Expected Solution Participants are expected to develop a scalable, secure, interoperable and user-friendly APAAR-based Student Data Management and Verification System that can be integrated with the AICTE portal. The solution should provide: Institution → Student Data Upload → APAAR Validation → Data Verification → AICTE Central Repository → Analytics & Monitoring → Authorized Verification Expected Outcome The proposed solution should enable AICTE to establish authentic and verifiable student records linked to a national academic identity, facilitate real-time monitoring

of admissions, detect excess enrolment, and strengthen data-driven governance and policy formulation. Innovation Expected The solution should go beyond simple data collection by providing automated validation, intelligent anomaly detection, real-time monitoring, secure verification, and analytics, thereby creating a reliable national-level student data ecosystem for AICTE-approved institutions.

Claude

The full text changes the picture. Requirements 5, 6, 8 and 10 (verification for outside stakeholders, verifiable records, consent, APAAR ID on certificates) bring part of your v2 trust-layer work back into scope. I'll update the report to reflect that.

Updating the opening sections and the requirement tables first.

Now the middle sections: student attributes, anomaly detection, real-time integration, API and reports.

Now the trust, verification and consent section, then the plan, demo and open questions.

I updated the report to match the full problem statement. Requirements 5 to 10 change the plan, and my first version had left out part of what they ask for.

What changed

- **Requirements 5 to 10:** these ask for secure verification for AICTE, institutions, employers and regulators, and for verifiable student records. They also ask for consent and audit trails, reports, and the APAAR ID printed on certificates. I had parked the signed-record and selective-disclosure work from your v2 as "roadmap only". It is now a core module, narrowed to enrolment and identity records rather than marksheets.
- **Scope decisions:** forensics, OCR and credit transfer stay out because nothing in the statement asks for them.
- **Verification (section 10):** a bare APAAR ID never returns data, since it will be printed on certificates. Each verifier sees only what its role and the student's consent allow, as a signed record with a tick-by-tick verification receipt.
- **Certificates:** a certificate stamp that returns the APAAR ID next to the student ID plus a QR token, and a check on certificate lists. Certificates come from degree-awarding bodies, mostly universities, so the plan covers them too.
- **Innovation clause:** anomaly detection is now layered and explainable instead of a bare Isolation Forest. It uses peer benchmarking, change-point detection, outlier scoring with reasons, and a

fixed review budget. Dashboards update live.

- **Integration and reports:** the integration approach (OpenAPI, signed webhooks, SFTP drop, adapters behind interfaces) and a report catalog with snapshot IDs and small-count suppression are both in the doc.
- **Plan and demo:** the 36-hour plan, team roles, demo (now 5 minutes), idea PPT outline and open questions are all updated. The plan now has a cut line for what to drop if you fall behind.

Still unconfirmed

- Whether you can get ABC/APAAR verification API access for a prototype. If not, the demo uses a mock and says so.
- The NRI cap and the vacant-seat clause wording, which should be checked against the handbook text.
- The SIH rule on pre-built code.

The full report is at https://claude.ai/artifact/5Hd3tienC9ipwJWXPgfNUu. If you want to continue, I can write the rules specification, database schema and OpenAPI file next.

2026-10-26 20:05:10 **You**

i dont think so i will be getting verified api access keep it with mock data only on which we will make it further lets just proceed with the project making i want u to act as senior and lead architect of the project and prepare a work flow architecture and whole working engine for the project how will we proceed and work professioanally and develop something which will genuinely be solving the problem do it how so ever u think w can work smoothly and efficiently keep in mind i am the only one working on it but this shouldnt compromise on the level and quality of the project i want this to be one of the best projects in my resume and industry level project doesnt matter how complex it is i will work on it no matter what i need to learn and what not keep the process non negotiable be a strict head who wants best work to be done also keep the process and the project flexible so as if we need to change or add anything in between the development we easily can

Claude

Plan: I'll build the real engine core first (rules engine, validation pipeline, mock APAAR registry, trust layer, synthetic data with seeded defects, evaluation harness, tests), prove it runs, then write the playbook around what exists. Checking the environment first.

2026-10-27 09:01:09 **You**

Continue

Claude

The worked examples A–E and the lifecycle work. Now the real test: the seeded-defect evaluation.

The engine passes 37 tests and processes about 145k rows in 7 seconds. Next I'll add the process files (README, ADRs, assumptions register, CI) and package everything.

The engine is packaged. Now the playbook that explains how it works and how we proceed, written against what actually exists and was measured.

文件：apaar-core.zip

I built the working engine core (milestone M0) and wrote the playbook for everything after it. Everything below was measured in my sandbox, so re-run it on your machine.

APAAR verification uses a mock behind an interface, as you decided. A real adapter can replace it later without touching the validation code.

What the engine does today

- **Rules as data:** 32 rules live in `rules/ay_2026_27.yaml`, each with a severity, a handbook clause reference and fix text. Changing a cap or a severity is a YAML edit plus a test.
- **Validation:** it checks schema, APAAR IDs, duplicates, capacity and cross-field logic. It also reconciles declared totals against the student rows and detects cross-institution overlap.
- **Over-admission handling:** it implements your worked cases A to E, including the vacant-seat exception. Case D is a warning rather than a false alarm.
- **Cross-institution privacy:** an institution sees only "another active enrolment exists", while AICTE sees the full case.
- **Submission workflow:** a second person must approve certification. The result is a signed receipt, and tampering with the audit log or the receipt is detected.

- **Tests:** 37 tests pass. The engine validates about 145,000 rows in 7 seconds (about 21,000 rows per second, single thread, in memory).

Read the 1.00 accuracy score with care. I wrote both the data generator and the rules, so a perfect score only shows the engine does what I specified. It does not show the rules match AICTE's handbook. Several cap values come from news reports and secondary sources, and the assumptions register lists each one for you to verify.

The playbook covers the workflow, the process rules I will enforce, milestones M1 to M8 with exit criteria and hours, the learning order, a risk register and a cut line for the finale. The rules that keep the project flexible are in the repo's `docs/PROCESS.md` .

Check the SIH rulebook on pre-built code this week. If the prototype has to be built during the 36-hour finale, this repo is still a strong resume project, but the finale plan changes.

Next steps

1. Unzip the repo and run `pip install -e ".[dev]"` , `pytest` , `apaar demo` and `apaar eval` . Tell me what you see.
2. Tell me how many hours a week you can commit, so we can pick between the six-week finale plan and the full build.
3. Then we start M1: the API, the database and tenant isolation.

Files are in /mnt/user-data/outputs:

- apaar-core.zip

The playbook is at https://claude.ai/artifact/7uXAoVRrpjZhy694U4kPQK.

2026-10-28 18:47:52 **You**

Okay, so basically what I'm telling you right now is exactly how you have to think and exactly how you have to work. So basically there is this thing that I have to show my mentor that what actually my project is all about and what I am thinking about it and how it should look like. And I want you to design it. Like, I want a basically a script, proper workflow, proper architecture of this project. I'm gonna work on this project very deeply and very thoroughly to make this work like more professionally and make it a successful one for my resume as well. Make it a standard project. Like, so much of industry level things in this that I get to learn things by building this,

and also that my mentor would like it. So basically tomorrow I need to show my mentor that what my project is actually all about, what the problem statement is, and what solution I have come to, and what is the tech stack and all the related things to the project, that I have to show it to him such that he makes my registration and I get the entry in the competition. But apart from the competition, I want this project to be a great one, actually the one that stands somewhere and definitely a winning project as well. So you have to work accordingly and make an architecture, workflow, proper report, proper structure, and everything important of this project so that I can use it and at least make something to show it to him tomorrow, explain it to him tomorrow that what my project actually is, and it actually solves a problem, and he gets convinced that yes, we can actually win this program and we can actually win this competition and do something.

Claude

Tomorrow's pack will be a short mentor deck to present, plus a briefing document with your speaking script and likely mentor questions. First I'll check which deck format is available.

2026-10-28 19:10:54 **You**

Continue

Claude

Your mentor pack is ready: a deck to present and a briefing document to rehearse from.

- **Deck:** 7 slides covering the problem, the solution, the architecture, what makes it different, the tech stack, and your six-week plan and what you need from your mentor. It is at https://claude.ai/artifact/HaGZaY98FHnAZmyyWNTfKn. You can download it as PowerPoint or PDF from there. I did not preview it visually, so click through it once before you present.
- **Briefing document:** at https://claude.ai/artifact/G98axXDEyd6XYRFeJnmjz7. It has an 8-minute spoken script matched to the slides, a 30-second pitch, a plain-words map of the ten requirements, answers to 11 likely mentor questions, and the limits to state yourself.

Run `apaar demo` live from the repo for about a minute. The main lines to point at are case D showing a warning instead of a false alarm, self-approval being refused, and a tampered audit log failing verification.

Things to be careful about

- **The numbers on slide 5 have limits.** They are 32 rules, 7 layers, 37 tests and about 21,000 rows per second. They come from synthetic data with planted defects and a mock APAAR registry, and both the deck and the script say so.
- **Team size.** Secondary sources say the hackathon needs a team of six including at least one woman. You told me you are working alone, so confirm the rule with your mentor tomorrow. The briefing lists the roles you would need.
- **Pre-built code.** Ask whether code written before the hackathon is allowed. If it is not, the working engine becomes preparation, and the plan changes.
- **Unverified rule values.** Some values, such as the supernumerary percentages and the NRI cap, come from secondary sources, so please don't present them as AICTE's official figures.

Tell me what your mentor says tomorrow, and especially the team and deadline details. Then we start the first build milestone: the API, the database and tenant isolation.

2026-10-28 19:23:16 **You**

okay now give me complete workflow architecture structure of the project along with the detailed tech stacksuch that i start working on it without wasting any more tim we gotta develop something really strong and way more powerful and useful to show such that it actually solves the problem we are asked to solve and put our ideas into it also keep it completely flexible and not rigid such that we can make the changes if required at the time of developing i dont care how much of the concept being used is complex i am ready to learn it and implement it i just want everything detailed and completely structured with the commands ideas changes and everything in the report i dont care how much time it takes it should be modernely developed and a fully optimised system also i want a report as well where in everything is explained how things will go and everything else along with that a script for the project and a md file as well in which the whole project along with the problem we are solving is listed in depth such that anyone who reads it understands what we are making how what problem is it solving how is it different and

unique from different solutions and etc topics which are necessary in that md file work on it and do keep in mind u have to act like a lead architect and a lead software engineer and a developer and have to act skillfully like u r the one leading this project go on and start working on it

Claude

I'll lead this in three parts. First I'll start the build with the API layer and a real PostgreSQL tenant-isolation test, then write the project file and demo script, then the full architecture report. Starting with what I can verify here.

Embedded PostgreSQL 16 works here, so I can test row-level security against a real database instead of describing it. Building the persistence layer first.